THAPAR UNIVERSITY

Electronics and Computer Engineering

Energy Consumption Analysis

Using PCA and K-Means Clustering

PROJECT REPORT

Submitted by

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Project Links

Energy Consumption Project Website

Energy-Consumption-PCA-KMeans GitHub Repository

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Abstract

Energy consumption data can help us understand usage patterns and identify unusual observations. This project analyzes weekly energy consumption data using Exploratory Data Analysis (EDA), Principal Component Analysis (PCA), and K-Means clustering.

The reported dataset contains 321 records and 156 weekly measurement columns. PCA is used to reduce the number of dimensions, while K-Means groups similar records.

The first two principal components explain approximately 99.9% of the variance in the reported analysis. Using three clusters, K-Means produces a silhouette score of 0.957. However, the cluster sizes are highly uneven, indicating that extreme observations may strongly influence the results.

The project demonstrates the use of unsupervised machine learning to explore data and understand its underlying structure.

1 Introduction

Energy consumption varies over time and between records. Analyzing these variations helps us understand patterns, relationships, and unusual values within a dataset.

This project combines data visualization, dimensionality reduction, and clustering to study weekly energy measurements. Rather than predicting future consumption, the main aim is to explore the data and identify groups of similar records.

1.1 Objectives

- Understand and prepare the dataset.
- Perform exploratory data analysis.
- Apply PCA to reduce dimensionality.
- Use K-Means to group similar records.
- Select the number of clusters using the elbow method.
- Evaluate clustering using the silhouette score.
- Interpret results and identify limitations.

2 Dataset and Preparation

The reported dataset contains 321 records and 156 weekly measurement columns. The weekly measurements are used to study consumption patterns and similarities between records.

Data preparation includes converting measurements to numeric values and handling missing data. In the reported workflow, missing values are filled using the median of the corresponding column.

Standardization is used to place features on a comparable scale. For a feature x, the standardized value is:

$$z = \frac{x - \mu}{\sigma}$$

where μ is the mean and σ is the standard deviation.

The dataset's original source should be verified using its documentation before submission.

3 Methodology

3.1 Exploratory Data Analysis

EDA helps us understand the distribution of consumption, weekly trends, correlations, and possible outliers. Graphs make these characteristics easier to interpret before applying machine learning algorithms.

3.2 Principal Component Analysis

PCA transforms the original features into a smaller number of principal components while preserving as much variation as possible.

The first two components are used to visualize the data in two dimensions, making it easier to examine its structure.

3.3 K-Means Clustering

K-Means is an unsupervised learning algorithm that groups records according to their similarity. It assigns records to clusters based on their distance from cluster centroids.

The objective is to minimize the within-cluster sum of squared distances:

$$J = \sum_{i=1}^{K} \sum_{x \in C_i} \|x - \mu_i\|^2$$

3.4 Elbow Method

The elbow method compares the clustering inertia for different values of K. The point where the reduction in inertia begins to slow down helps select the number of clusters. The reported analysis selects $K = 3$.

3.5 Silhouette Score

The silhouette score measures how well records fit within their assigned clusters compared with other clusters. A score closer to 1 generally indicates stronger separation. However, it should be interpreted alongside cluster sizes and the presence of extreme observations.

4 Exploratory Data Analysis and Visualizations

4.1 Distribution of Average Consumption

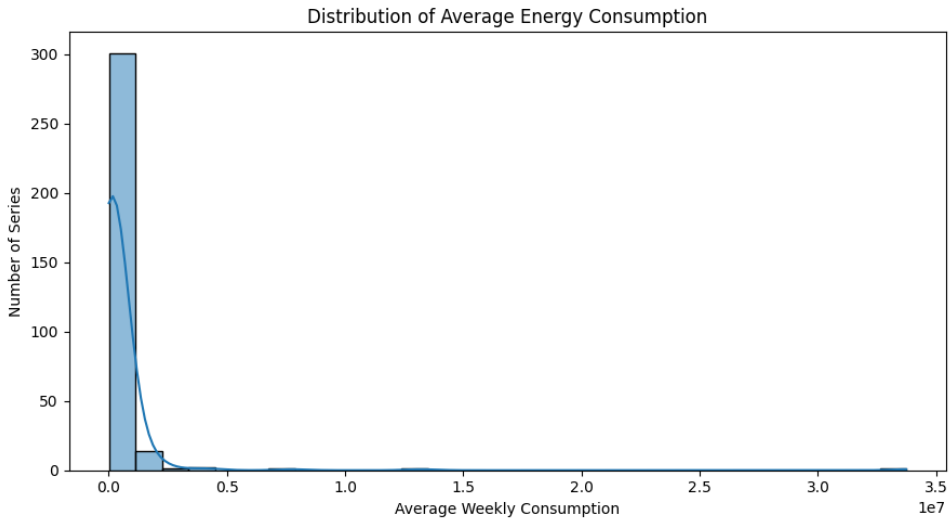


Figure 1: Distribution of average energy consumption

Explanation: The histogram shows the distribution of average consumption. The reported data contains many relatively small values and a few unusually large observations.

4.2 Weekly Consumption Trend

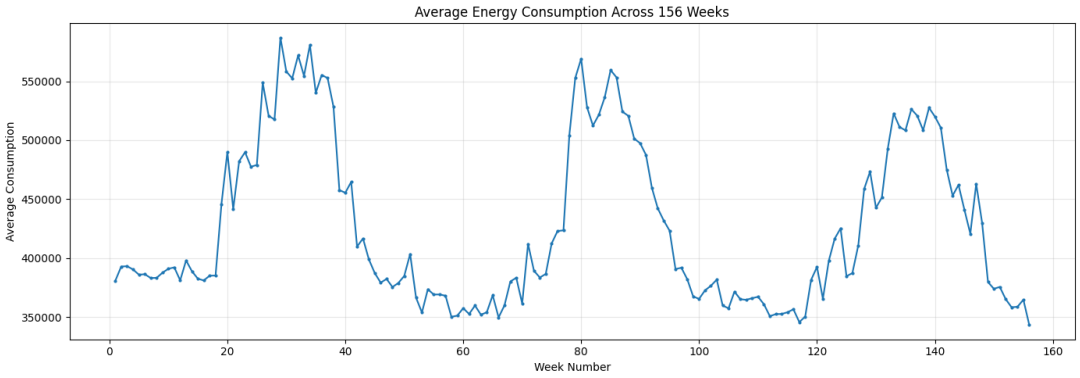


Figure 2: Average energy consumption across weeks

Explanation: This graph shows how average consumption changes over time. Peaks were reported around weeks 30, 80, and 135.

4.3 Box Plot

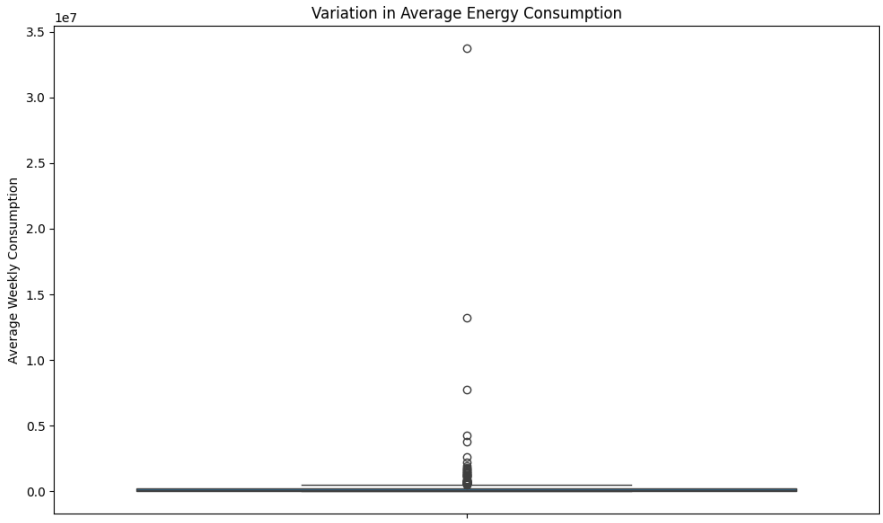


Figure 3: Spread of average energy consumption

Explanation: The box plot displays the spread of consumption and highlights potential outliers that may influence later analysis.

4.4 Correlation Matrix

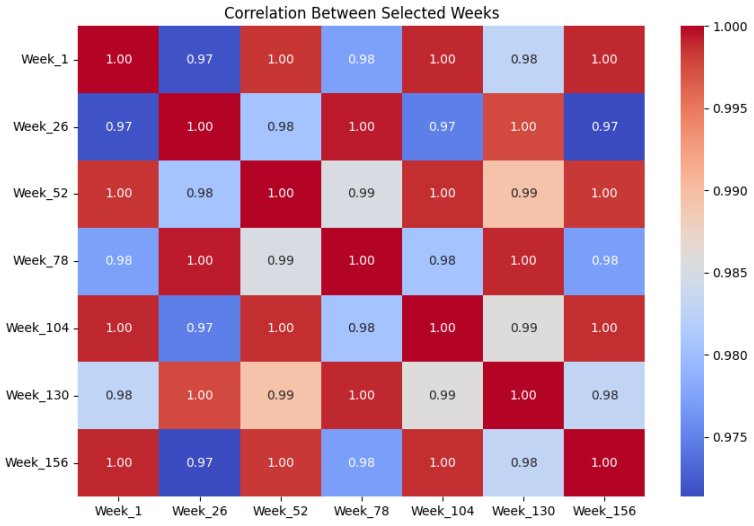


Figure 4: Correlation between selected weekly measurements

Explanation: The reported correlations range from approximately 0.97 to 1.00, indicating strong positive relationships between the selected weekly measurements.

5 PCA and Clustering Results

5.1 PCA Projection

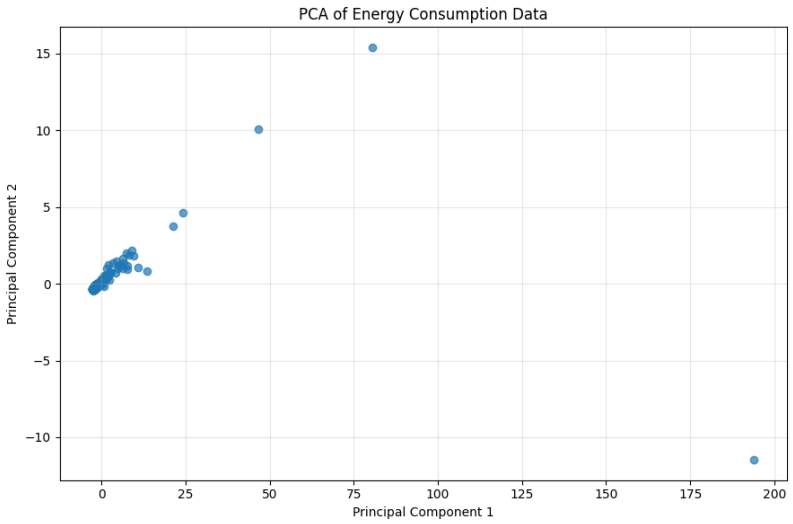


Figure 5: Data projected onto the first two principal components

Explanation: The first two components explain approximately 99.9% of the variance in the reported analysis, allowing the records to be visualized in two dimensions.

5.2 Elbow Method

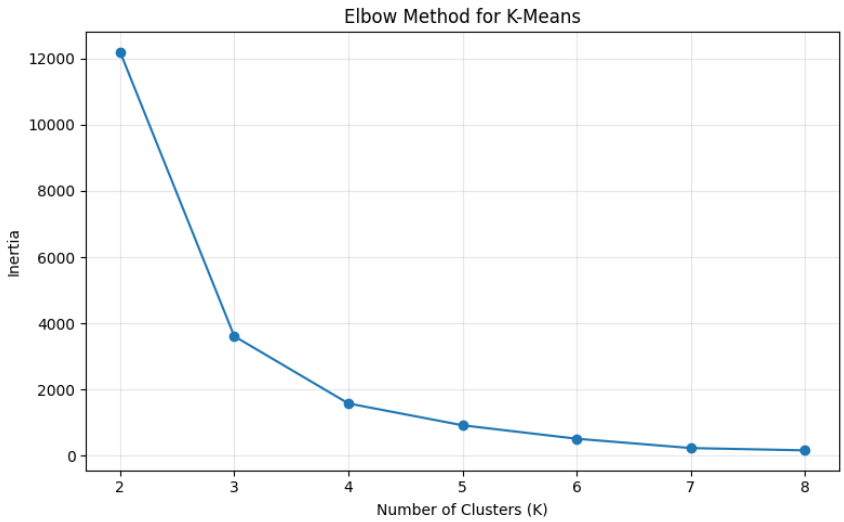


Figure 6: Elbow plot for selecting the number of clusters

Explanation: The reported elbow plot supports choosing three clusters based on the observed change in inertia.

5.3 K-Means Clustering

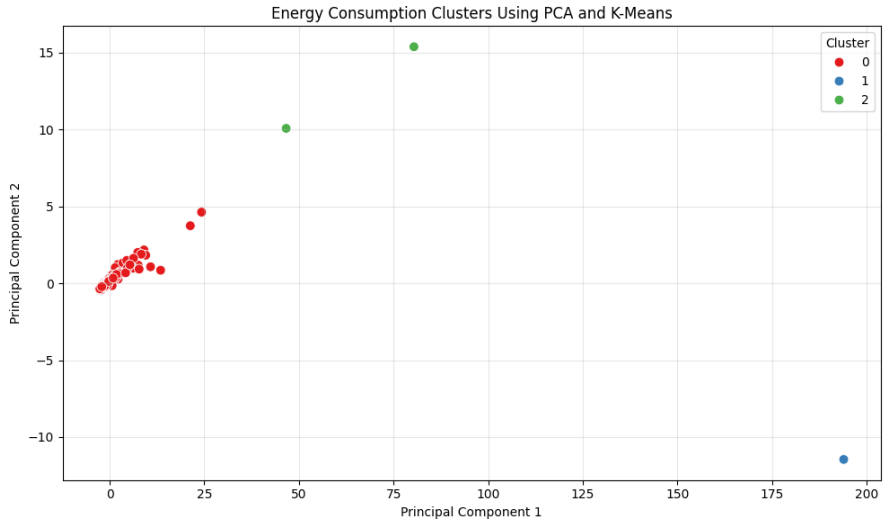


Figure 7: K-Means clusters visualized using PCA

Explanation: The reported result contains one large cluster and two very small clusters. This suggests that extreme observations strongly influence the grouping.

5.4 Mean Consumption by Cluster

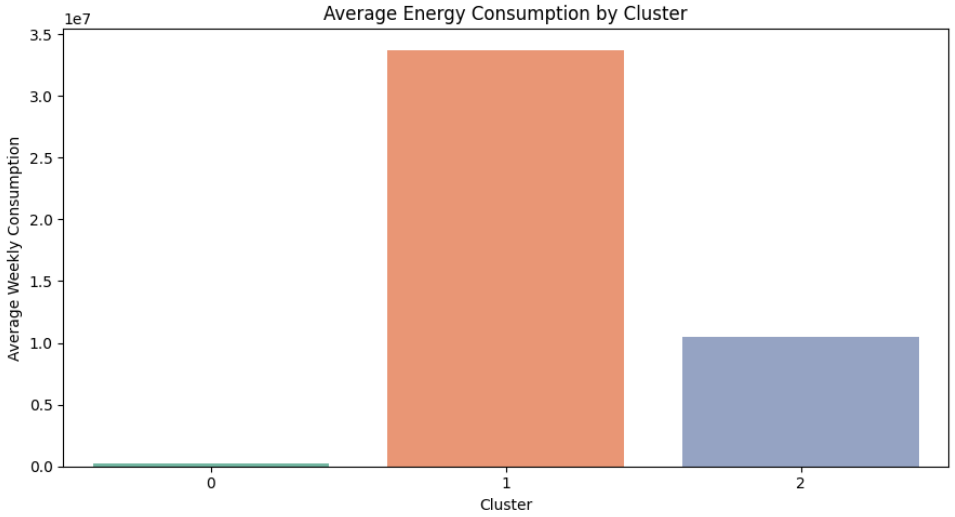


Figure 8: Mean average consumption for each cluster

Explanation: This graph compares consumption levels across the clusters. The reported analysis shows substantial differences between the cluster means.

6 Discussion

The analysis combines EDA, PCA, and K-Means to explore weekly energy consumption data. PCA provides a lower-dimensional view, while K-Means identifies groups of records with similar characteristics.

The reported silhouette score is 0.957. However, most records belong to one cluster, while the other two clusters contain only one and two records, respectively.

This uneven distribution is an important limitation. A high silhouette score does not necessarily mean that the clusters are balanced or practically useful. Extreme observations can strongly affect distance-based algorithms.

Future work could investigate logarithmic transformation, examine extreme records separately, and compare alternative clustering configurations.

7 Proposed Optimization Techniques

The proposed optimization strategy focuses on identifying abnormal consumption patterns, analyzing peak energy usage, and improving the reliability of clustering results. Extreme energy measurements will be investigated to distinguish genuine high consumption from potential data anomalies. Weekly consumption trends will be examined to identify recurring peaks that may indicate opportunities for improved energy management. Principal Component Analysis (PCA) will be used to reduce the dimensionality of the dataset while preserving most of its variance, and K-Means clustering will group similar time series based on their consumption patterns. Since extreme values can strongly influence clustering, suitable data transformations, such as logarithmic scaling for non-negative measurements, will be evaluated to improve cluster separation. The resulting clusters will be assessed using silhouette scores, cluster sizes, and consumption profiles. These analyses can help identify unusual consumption patterns and suggest targeted monitoring and energy-saving measures, subject to further investigation of the underlying causes.

8 Conclusion

This project demonstrates how exploratory data analysis, PCA, and K-Means can be used to investigate weekly energy consumption data.

PCA reduces the dimensionality of the dataset, and K-Means provides an initial grouping of records. The visualizations help explain the dataset's characteristics and the resulting clusters.

An important lesson from this project is that machine learning results must be interpreted

carefully. Although the reported silhouette score is high, the uneven cluster sizes indicate that extreme observations have a strong influence on the result.

Further investigation of outliers and alternative preprocessing methods could help determine whether more informative clusters can be obtained.

9 References

1. K. Pearson, "On Lines and Planes of Closest Fit to Systems of Points in Space," 1901.
2. S. Lloyd, "Least Squares Quantization in PCM," 1982.
3. Scikit-learn documentation: https://scikit-learn.org/
4. Pandas documentation: https://pandas.pydata.org/docs/